

FitCore AI — Full Product Requirements Document (PRD)

Document control

Field	Value
Product	FitCore AI Gym Management Platform
Version	1.0
Type	Multi-tenant B2B SaaS
Primary DB	PostgreSQL
Auth	Firebase Authentication (primary)
Storage	AWS S3

1. Product overview
- 1.1 Vision
- A cloud gym-management platform that automates operations, centralizes members across branches, integrates physical access (QR, RFID, fingerprint), and delivers predictive business insights for owners and franchises.
- 1.2 Target customers
- Independent gyms
 - Multi-branch fitness centers
 - Franchises
 - PT studios, yoga, CrossFit, martial arts

1.3 Success metrics

Metric	Target (12 months post-launch)
Admin task time	−40% vs manual ops
Member check-in time	< 3 seconds (QR/fingerprint)
Payment collection on time	+25%
Churn identification	Flag high-risk members 14+ days early

Platform uptime	99.5%
Concurrent branches per tenant	50+

1.4 Tenancy model (decision)

- **SaaS multi-tenant:** one deployment serves many gym **organizations**.
- **Organization** = billing customer (gym brand / franchise).
- **Branch** = physical location under one organization.
- **Platform Super Admin** = FitCore operator (all orgs).
- **Org Owner / Org Admin** = customer admin (all branches in their org, unless scoped).

2. Personas & roles

Persona	Goals	Primary surfaces
Platform Super Admin	Operate SaaS, support tenants	Platform console
Org Owner	Revenue, growth, all branches	Admin dashboard
Branch Admin	Day-to-day branch ops	Branch dashboard
Trainer	Client progress, attendance	Trainer dashboard
Receptionist	Check-in, payments, registration	Front desk
Member	Access, workouts, renewals	Mobile / web member portal

2.1 RBAC matrix (summary)

Permission	Platform Super Admin	Org Admin	Branch Admin	Trainer	Receptionist	Member
Manage organizations	✓	—	—	—	—	—
Manage all branches (org)	✓	✓	—	—	—	—
Manage assigned branch	✓	✓	✓	—	—	—

Financial analytics (org)	✓	✓	—	—	—	—
Branch revenue view	✓	✓	✓	—	—	—
Member CRUD	✓	✓	✓	—	—	—
Delete member	✓	✓	✓	—	—	—
Approve registration	✓	✓	✓	—	✓	—
Check-in / manual attendance	✓	✓	✓	—	✓	—
Assigned members only	—	—	—	✓	—	—
Workout plans / notes	✓	✓	✓	✓	—	View own
Device / access config	✓	✓	✓	—	—	—
AI insights (org/branch)	✓	✓	✓	—	—	—
Personal AI fitness assistant	—	—	—	—	—	✓
Audit logs	✓	✓ (org)	✓ (branch)	—	—	—
Export reports	✓	✓	✓	—	—	—

Billing / subscription (SaaS)	✓	✓	—	—	—	—
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Roles are assigned **per organization** and optionally **scoped to branch** (user_branch_roles).

3. Functional requirements

3.1 Dashboard & analytics

FR-DASH-01 Real-time KPIs: total members, active memberships, MTD revenue, today attendance, expiring (7/30 days), new registrations, peak hour (today).

FR-DASH-02 Trends: attendance (7/30/90d), revenue, membership growth.

FR-DASH-03 Branch comparison (org admin+).

FR-DASH-04 Export snapshot (PDF/CSV) per widget where applicable.

AI (Phase 2+)

FR-AI-01 Churn risk score 0–100 per member.

FR-AI-02 Peak hour prediction per branch/day-of-week.

FR-AI-03 Revenue forecast (30/90d).

FR-AI-04 Renewal likelihood.

MVP substitute: inactive days, overdue payments, expiring membership lists.

3.2 Member management

FR-MEM-01 CRUD member profile with branch assignment.

FR-MEM-02 Membership plan attach, upgrade/downgrade, proration rules (configurable).

FR-MEM-03 Freeze / unfreeze with reason and date range.

FR-MEM-04 Auto-renewal flag per membership.

FR-MEM-05 Body measurements and progress timeline.

FR-MEM-06 Medical notes and emergency contacts (restricted visibility).

FR-MEM-07 QR access ID generation and rotation.

FR-MEM-08 Trainer assignment (optional, many-to-many allowed).

FR-MEM-09 Multi-branch membership (same org, multiple branch access flags).

3.3 Self-registration

FR-REG-01 Public URL: https://{org_slug}.fitcore.app/register or custom domain.

FR-REG-02 QR deep link to same flow.

FR-REG-03 Required fields: name, mobile, email, gender, DOB, branch, plan.

FR-REG-04 Optional: photo, medical, goals, emergency contact.

FR-REG-05 OTP (SMS/email), CAPTCHA, rate limit (e.g. 5/hour/IP).

FR-REG-06 Status flow: draft → otp_verified → pending_approval → approved → active / rejected.

FR-REG-07 On approval: create user (if needed), member, membership, QR, notify member.

3.4 Access control

FR-ACC-01 Methods: QR (Phase 1), manual (Phase 1), RFID (Phase 2), fingerprint ZKTeco (Phase 2), face (Phase 3).

FR-ACC-02 Deny access if membership not active or frozen.

FR-ACC-03 Optional: anti-passback (one check-in per X minutes).

FR-ACC-04 Audit every access attempt (granted/denied).

3.5 ZKTeco integration (Phase 2)

FR-ZK-01 Register devices per branch (IP, model, serial).

FR-ZK-02 Edge agent on LAN syncs attendance logs to API.

FR-ZK-03 Map device_user_id ↔ member_id.

FR-ZK-04 Enrollment workflow at front desk after approval.

FR-ZK-05 No raw fingerprint images in cloud DB; templates on device only.

3.6 Attendance

FR-ATT-01 Record check-in with source enum.

FR-ATT-02 Optional check-out.

FR-ATT-03 Trainer attendance (staff).

FR-ATT-04 Heatmaps and peak analysis.

FR-ATT-05 Realtime dashboard updates via WebSocket/FCM.

3.7 Payments & billing

FR-PAY-01 Membership invoices (line items, tax, discount).

FR-PAY-02 Payment methods: card (gateway), cash, bank transfer, UPI (region-specific).

FR-PAY-03 Recurring billing via gateway subscriptions.

FR-PAY-04 Reminders: T-7, T-3, T-0, overdue (configurable).

FR-PAY-05 Channels: email (MVP), push (Phase 2), SMS/WhatsApp (Phase 2).

FR-PAY-06 Failed payment retry and staff alert.

Member billing vs SaaS subscription (org pays FitCore): separate tables (subscriptions vs memberships).

3.8 Notifications

- FR-NOT-01** Templates per event type and channel.
- FR-NOT-02** Delivery log with status (queued, sent, failed).
- FR-NOT-03** Member preference opt-out (marketing vs transactional).

3.9 Mobile app (Phase 2)

Digital card, QR, attendance history, workouts, push, renewals, AI fitness assistant (workout/nutrition tips, engagement nudges).

3.10 Reports

Revenue, attendance, trainer, membership, branch performance, churn. Formats: PDF, CSV, XLSX. Async generation for large reports.

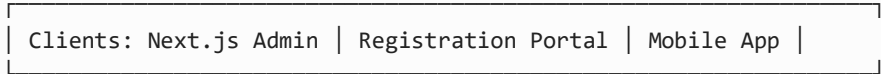
3.11 Multi-branch

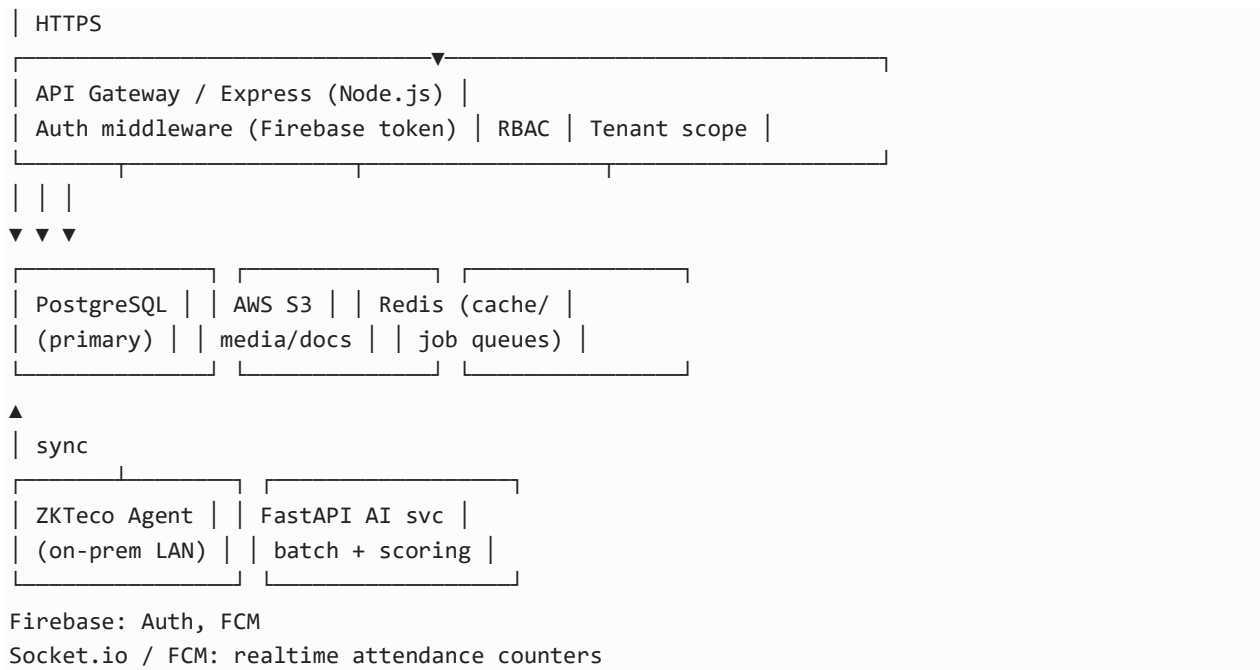
Branch switcher in UI; all queries scoped by `organization_id` + permitted `branch_id(s)`. Shared membership across branches (org-level plan flag).

4. Non-functional requirements

ID	Requirement
NFR-01	HTTPS only; TLS 1.2+
NFR-02	500–5000+ members per org; horizontal API scaling
NFR-03	API p95 < 500ms for CRUD; dashboard aggregates < 2s (cached)
NFR-04	Daily DB backups; RPO ≤ 24h, RTO ≤ 4h (target)
NFR-05	Audit log for sensitive actions
NFR-06	GDPR-style export/delete for member PII (process)
NFR-07	Rate limiting on auth and registration
NFR-08	Idempotent payment webhooks

5. System architecture





6. Authentication & authorization

- **Identity provider:** Firebase Authentication (email/password, phone OTP optional).
- **API:** Verify Firebase ID token; map `firebase_uid` → users.
- **Authorization:** JWT custom claims or DB lookup: `organization_id`, `roles[]`, `branch_ids[]`.
- **Do not store** `password_hash` in PostgreSQL when using Firebase.
- **Service accounts:** Edge agent and webhooks use API keys + HMAC.

7. Integrations

Integration	Phase	Purpose
Firebase Auth	1	Login
FCM	2	Push
Stripe/Razorpay (TBD)	1	Payments
SendGrid/SES	1	Email
Twilio/MessageBird	2	SMS
WhatsApp Business API	2	WhatsApp
ZKTeco SDK	2	Devices

8. User flows (condensed)

Registration: Form → CAPTCHA → OTP → registration_requests.pending → Admin approve → Member + Membership + QR email.

Check-in (QR): Scan → validate membership → access_events → dashboard realtime.

Fingerprint: Device punch → agent sync → dedupe → attendance → dashboard.

9. API modules (outline)

- /v1/auth — session, profile, roles
- /v1/organizations, /v1/branches
- /v1/users, /v1/roles
- /v1/members, /v1/memberships, /v1/plans
- /v1/registrations
- /v1/attendance, /v1/access-events
- /v1/devices, /v1/device-sync
- /v1/invoices, /v1/payments, /v1/reminders
- /v1/notifications
- /v1/analytics, /v1/reports
- /v1/ai/scores
- /v1/audit-logs

10. Development phases

Phase	Scope	Duration (indicative)
1 – MVP	Single org multi-branch optional; auth; members; plans; memberships; QR + manual attendance; invoices; email reminders; basic dashboard; registration + approval	10–14 weeks
2	ZKTeco agent; RFID; mobile app; trainer portal; push; SMS; rule-based AI	8–12 weeks
3	ML churn/forecast; face recognition; franchise billing; advanced CRM	12+ weeks

11. Out of scope (v1)

- Face recognition
- Wearables / smart lockers
- Full CRM marketing automation
- Native franchise royalty engine (Phase 3)

12. Acceptance criteria (MVP sample)

1. Org admin can create branch, plan, member, and active membership.
2. Member receives QR; scan at reception records attendance within 2s.
3. Expired member scan returns denied with reason.
4. Public registration completes OTP and appears in pending queue.
5. Branch admin approval activates membership and sends email.
6. Dashboard shows today's attendance and MTD revenue matching DB.
7. Receptionist cannot access org-wide revenue API (403).

Full ERD — FitCore AI

ERD diagram (logical)

organizations(no attributes)branches(no attributes)membership_plans(no attributes)users(no attributes)members(no attributes)devices(no attributes)saas_subscriptions(no attributes)staff_schedules(no attributes)user_organization_roles(no attributes)user_branch_roles(no attributes)trainers(no attributes)memberships(no attributes)attendance_records(no attributes)access_events(no attributes)body_measurements(no attributes)member_trainers(no attributes)device_user_mappings(no attributes)member_qr_tokens(no attributes)registration_requests(no attributes)invoices(no attributes)payments(no attributes)payment_reminders(no attributes)device_sync_logs(no attributes)workout_plans(no attributes)workout_sessions(no attributes)workout_exercises(no attributes)ai_member_scores(no attributes)ai_branch_forecasts(no attributes)notifications(no attributes)notification_templates(no attributes)audit_logs(no attributes)hasoffersemploysownsowns billed_byassignedinstalledschedulesmay_behasscopedprofileh asgeneratesgeneratestracksassignedenrolledhasfrominstantiatedbilledpaid_bytriggersmapssyncsreco rdstrainscreatesreceivescontainscontainsscoredforecastreceivesusesperforms

Enum types

```
-- Role (platform + org)
CREATE TYPE platform_role AS ENUM ('platform_super_admin');
CREATE TYPE org_role AS ENUM ('org_owner', 'org_admin', 'branch_admin', 'trainer',
'receptionist');
-- Member
CREATE TYPE member_status AS ENUM ('pending', 'active', 'frozen', 'expired', 'cancelled',
'suspended');
CREATE TYPE gender_type AS ENUM ('male', 'female', 'other', 'prefer_not_to_say');
-- Registration
CREATE TYPE registration_status AS ENUM (
'draft', 'otp_sent', 'otp_verified', 'pending_approval', 'approved', 'rejected', 'expired'
);
-- Membership
CREATE TYPE membership_status AS ENUM ('pending', 'active', 'frozen', 'expired', 'cancelled');
CREATE TYPE billing_period AS ENUM ('daily', 'weekly', 'monthly', 'quarterly', 'yearly',
'one_time');
-- Attendance / Access
CREATE TYPE attendance_source AS ENUM ('qr', 'rfid', 'fingerprint', 'face', 'manual', 'api');
CREATE TYPE access_result AS ENUM ('granted', 'denied');
-- Payments
CREATE TYPE invoice_status AS ENUM ('draft', 'open', 'paid', 'partially_paid', 'void',
'overdue');
CREATE TYPE payment_status AS ENUM ('pending', 'completed', 'failed', 'refunded');
CREATE TYPE payment_method AS ENUM ('card', 'cash', 'bank_transfer', 'upi', 'wallet',
'other');
-- Devices
CREATE TYPE device_type AS ENUM ('zkteco_fingerprint', 'zkteco_face', 'rfid_reader',
'qr_scanner');
CREATE TYPE device_status AS ENUM ('online', 'offline', 'maintenance', 'decommissioned');
-- Notifications
CREATE TYPE notification_channel AS ENUM ('email', 'sms', 'push', 'whatsapp', 'in_app');
CREATE TYPE notification_status AS ENUM ('queued', 'sent', 'delivered', 'failed', 'read');
-- SaaS
CREATE TYPE saas_plan_tier AS ENUM ('starter', 'growth', 'enterprise');
CREATE TYPE saas_subscription_status AS ENUM ('trialing', 'active', 'past_due', 'cancelled');
```

Table definitions

Platform & tenancy

organizations

Column	Type	Constraints	Description
id	UUID	PK	

name	VARCHAR(255)	NOT NULL	Legal/display name
slug	VARCHAR(100)	UNIQUE, NOT NULL	Subdomain
logo_url	TEXT		S3 URL
email	VARCHAR(255)		Billing contact
phone	VARCHAR(50)		
timezone	VARCHAR(64)	DEFAULT 'UTC'	
currency	CHAR(3)	DEFAULT 'USD'	ISO 4217
settings	JSONB	DEFAULT '{}'	Tax, reminder days, features
status	VARCHAR(20)	DEFAULT 'active'	active, suspended
created_at	TIMESTAMPTZ	NOT NULL	
updated_at	TIMESTAMPTZ	NOT NULL	
deleted_at	TIMESTAMPTZ		Soft delete

branches

Column	Type	Constraints	Description
id	UUID	PK	
organization_id	UUID	FK → organizations, NOT NULL	
name	VARCHAR(255)	NOT NULL	
code	VARCHAR(50)		Short code
address_line1	VARCHAR(255)		
address_line2	VARCHAR(255)		

city	VARCHAR(100)		
state	VARCHAR(100)		
postal_code	VARCHAR(20)		
country	CHAR(2)		
contact_phone	VARCHAR(50)		
contact_email	VARCHAR(255)		
latitude	DECIMAL(10,7)		
longitude	DECIMAL(10,7)		
operating_hours	JSONB		Per day schedule
status	VARCHAR(20)	DEFAULT 'active'	
created_at	TIMESTAMPTZ	NOT NULL	
updated_at	TIMESTAMPTZ	NOT NULL	

Indexes: (organization_id), (organization_id, code) UNIQUE

Users & RBAC

users

Column	Type	Constraints	Description
id	UUID	PK	
firebase_uid	VARCHAR(128)	UNIQUE	Nullable for system users
email	VARCHAR(255)	NOT NULL	
phone	VARCHAR(50)		

full_name	VARCHAR(255)	NOT NULL	
avatar_url	TEXT		
is_active	BOOLEAN	DEFAULT true	
last_login_at	TIMESTAMPTZ		
created_at	TIMESTAMPTZ	NOT NULL	
updated_at	TIMESTAMPTZ	NOT NULL	

Note: No password_hash when using Firebase.

user_organization_roles

Column	Type	Constraints
id	UUID	PK
user_id	UUID	FK → users
organization_id	UUID	FK → organizations
role	org_role	NOT NULL
created_at	TIMESTAMPTZ	

UNIQUE: (user_id, organization_id, role)

user_branch_roles

Column	Type	Constraints
id	UUID	PK
user_id	UUID	FK → users
branch_id	UUID	FK → branches
role	org_role	NOT NULL
created_at	TIMESTAMPTZ	

UNIQUE: (user_id, branch_id, role)

platform_admins		
Column	Type	Constraints
user_id	UUID	PK, FK → users
created_at	TIMESTAMPTZ	

Members			
members			
Column	Type	Constraints	Description
id	UUID	PK	
organization_id	UUID	FK, NOT NULL	
branch_id	UUID	FK → branches	Primary branch
user_id	UUID	FK → users	NULL until portal activated
member_code	VARCHAR(50)	NOT NULL	Human-readable ID
full_name	VARCHAR(255)	NOT NULL	
email	VARCHAR(255)		
phone	VARCHAR(50)		
gender	gender_type		
date_of_birth	DATE		
photo_url	TEXT		
status	member_status	DEFAULT 'pending'	
emergency_contact_name	VARCHAR(255)		

emergency_contact_phone	VARCHAR(50)		
medical_notes	TEXT		Restricted
fitness_goals	TEXT		
rfid_card_id	VARCHAR(100)		
joined_at	DATE		
created_at	TIMESTAMPTZ		
updated_at	TIMESTAMPTZ		
deleted_at	TIMESTAMPTZ		

UNIQUE: (organization_id, member_code)

Indexes: (organization_id, branch_id, status), (email), (phone)

member_branch_access

Column	Type	Description
id	UUID	PK
member_id	UUID	FK
branch_id	UUID	FK
is_primary	BOOLEAN	
created_at	TIMESTAMPTZ	

UNIQUE: (member_id, branch_id) — multi-branch access

member_qr_tokens

Column	Type	Description
id	UUID	PK
member_id	UUID	FK
token_hash	VARCHAR(255)	Store hash only

expires_at	TIMESTAMPTZ	Optional rotation
is_active	BOOLEAN	
created_at	TIMESTAMPTZ	

Trainers

trainers

Column	Type	Constraints
id	UUID	PK
user_id	UUID	FK → users, UNIQUE
organization_id	UUID	FK
branch_id	UUID	FK
specialization	VARCHAR(255)	
bio	TEXT	
is_active	BOOLEAN	
created_at	TIMESTAMPTZ	

member_trainers

Column	Type	Constraints
id	UUID	PK
member_id	UUID	FK
trainer_id	UUID	FK
assigned_at	TIMESTAMPTZ	
ended_at	TIMESTAMPTZ	

UNIQUE: (member_id, trainer_id) where ended_at IS NULL (partial index)

trainer_notes		
Column	Type	Description
id	UUID	PK
member_id	UUID	FK
trainer_id	UUID	FK
note	TEXT	
created_at	TIMESTAMPTZ	

Plans & memberships		
membership_plans		
Column	Type	Description
id	UUID	PK
organization_id	UUID	FK
name	VARCHAR(255)	
description	TEXT	
price	DECIMAL(12,2)	
billing_period	billing_period	
duration_days	INT	For fixed-term
trial_days	INT	
tax_rate	DECIMAL(5,2)	
allows_multi_branch	BOOLEAN	
is_active	BOOLEAN	
created_at	TIMESTAMPTZ	

updated_at

TIMESTAMPTZ

memberships

Column	Type	Description
id	UUID	PK
member_id	UUID	FK
plan_id	UUID	FK → membership_plans
branch_id	UUID	FK
status	membership_status	
start_date	DATE	
end_date	DATE	
auto_renew	BOOLEAN	
frozen_from	DATE	
frozen_until	DATE	
freeze_reason	TEXT	
cancelled_at	TIMESTAMPTZ	
created_at	TIMESTAMPTZ	
updated_at	TIMESTAMPTZ	

Indexes: (member_id, status), (end_date) for expiry jobs

Registration

registration_requests

Column	Type	Description
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id	UUID	PK
organization_id	UUID	FK
branch_id	UUID	FK
plan_id	UUID	FK
status	registration_status	
full_name	VARCHAR(255)	
email	VARCHAR(255)	
phone	VARCHAR(50)	
gender	gender_type	
date_of_birth	DATE	
photo_url	TEXT	
medical_conditions	TEXT	
fitness_goals	TEXT	
emergency_contact_name	VARCHAR(255)	
emergency_contact_phone	VARCHAR(50)	
otp_hash	VARCHAR(255)	
otp_expires_at	TIMESTAMPTZ	
captcha_verified	BOOLEAN	
ip_address	INET	
reviewed_by	UUID	FK → users
reviewed_at	TIMESTAMPTZ	

rejection_reason	TEXT	
member_id	UUID	FK → members
created_at	TIMESTAMPTZ	
updated_at	TIMESTAMPTZ	

Attendance & access

attendance_records

Column	Type	Description
id	UUID	PK
organization_id	UUID	FK
branch_id	UUID	FK
member_id	UUID	FK
user_id	UUID	FK
device_id	UUID	FK → devices
check_in_at	TIMESTAMPTZ	NOT NULL
check_out_at	TIMESTAMPTZ	
source	attendance_source	
external_punch_id	VARCHAR(100)	Dedupe from device
created_at	TIMESTAMPTZ	

UNIQUE: (device_id, external_punch_id) where not null

Indexes: (branch_id, check_in_at), (member_id, check_in_at)

access_events

Column	Type	Description
id	UUID	PK
organization_id	UUID	FK

branch_id	UUID	FK
member_id	UUID	FK
device_id	UUID	FK
source	attendance_source	
result	access_result	
deny_reason	VARCHAR(100)	expired, frozen, invalid_qr
occurred_at	TIMESTAMPTZ	

Devices (ZKTeco / RFID)

devices

Column	Type	Description
id	UUID	PK
organization_id	UUID	FK
branch_id	UUID	FK
name	VARCHAR(255)	
device_type	device_type	
manufacturer	VARCHAR(100)	e.g. ZKTeco
model	VARCHAR(100)	K40, F18
serial_number	VARCHAR(100)	UNIQUE
ip_address	INET	
port	INT	
api_key_hash	VARCHAR(255)	Edge agent auth
status	device_status	

last_sync_at	TIMESTAMPTZ	
settings	JSONB	
created_at	TIMESTAMPTZ	

device_user_mappings

Column	Type	Description
id	UUID	PK
device_id	UUID	FK
member_id	UUID	FK
device_user_id	VARCHAR(50)	ID on physical device
enrolled_at	TIMESTAMPTZ	
enrolled_by	UUID	FK → users
is_active	BOOLEAN	

UNIQUE: (device_id, device_user_id)

device_sync_logs

Column	Type	Description
id	UUID	PK
device_id	UUID	FK
sync_started_at	TIMESTAMPTZ	
sync_finished_at	TIMESTAMPTZ	
records_imported	INT	
status	VARCHAR(20)	success, failed
error_message	TEXT	

Payments

invoices

Column	Type	Description
id	UUID	PK
organization_id	UUID	FK
branch_id	UUID	FK
member_id	UUID	FK
membership_id	UUID	FK
invoice_number	VARCHAR(50)	UNIQUE per org
subtotal	DECIMAL(12,2)	
tax_amount	DECIMAL(12,2)	
discount_amount	DECIMAL(12,2)	
total_amount	DECIMAL(12,2)	
status	invoice_status	
due_date	DATE	
issued_at	TIMESTAMPTZ	
paid_at	TIMESTAMPTZ	
created_at	TIMESTAMPTZ	

invoice_line_items

Column	Type	Description
id	UUID	PK
invoice_id	UUID	FK
description	VARCHAR(255)	

quantity	INT	
unit_price	DECIMAL(12,2)	
amount	DECIMAL(12,2)	

payments

Column	Type	Description
id	UUID	PK
invoice_id	UUID	FK
member_id	UUID	FK
amount	DECIMAL(12,2)	
payment_method	payment_method	
status	payment_status	
gateway	VARCHAR(50)	stripe, razorpay
gateway_payment_id	VARCHAR(255)	
payment_date	TIMESTAMPTZ	
recorded_by	UUID	FK → users
metadata	JSONB	
created_at	TIMESTAMPTZ	

payment_reminders

Column	Type	Description
id	UUID	PK
invoice_id	UUID	FK
reminder_type	VARCHAR(20)	t_minus_7, overdue

scheduled_at	TIMESTAMPTZ	
sent_at	TIMESTAMPTZ	
channel	notification_channel	
status	notification_status	

Body & workouts

body_measurements

Column	Type	Description
id	UUID	PK
member_id	UUID	FK
recorded_at	TIMESTAMPTZ	
weight_kg	DECIMAL(5,2)	
height_cm	DECIMAL(5,2)	
body_fat_pct	DECIMAL(5,2)	
chest_cm	DECIMAL(5,2)	
waist_cm	DECIMAL(5,2)	
hips_cm	DECIMAL(5,2)	
recorded_by	UUID	FK → users
notes	TEXT	

workout_plans

Column	Type	Description
id	UUID	PK

member_id	UUID	FK
trainer_id	UUID	FK
title	VARCHAR(255)	
start_date	DATE	
end_date	DATE	
is_active	BOOLEAN	
created_at	TIMESTAMPTZ	

workout_sessions		
Column	Type	Description
id	UUID	PK
workout_plan_id	UUID	FK
day_label	VARCHAR(50)	
sort_order	INT	

workout_exercises		
Column	Type	Description
id	UUID	PK
workout_session_id	UUID	FK
exercise_name	VARCHAR(255)	
sets	INT	
reps	INT	
duration_sec	INT	
notes	TEXT	

workout_completions

Column	Type	Description
id	UUID	PK
member_id	UUID	FK
workout_session_id	UUID	FK
completed_at	TIMESTAMPTZ	

Notifications

notification_templates

Column	Type	Description
id	UUID	PK
organization_id	UUID	FK
event_type	VARCHAR(50)	membership_expiry, payment_due
channel	notification_channel	
subject	VARCHAR(255)	
body_template	TEXT	
is_active	BOOLEAN	

notifications

Column	Type	Description
id	UUID	PK
organization_id	UUID	FK
member_id	UUID	FK
user_id	UUID	FK
channel	notification_channel	

title	VARCHAR(255)	
body	TEXT	
status	notification_status	
sent_at	TIMESTAMPTZ	
read_at	TIMESTAMPTZ	
metadata	JSONB	
created_at	TIMESTAMPTZ	

member_notification_preferences		
Column	Type	Description
member_id	UUID	PK, FK
email_enabled	BOOLEAN	
sms_enabled	BOOLEAN	
push_enabled	BOOLEAN	
whatsapp_enabled	BOOLEAN	
marketing_opt_in	BOOLEAN	

AI & analytics		
ai_member_scores		
Column	Type	Description
id	UUID	PK
member_id	UUID	FK
score_type	VARCHAR(50)	churn, renewal

score	DECIMAL(5,2)	0-100
factors	JSONB	Explainability
calculated_at	TIMESTAMPTZ	

INDEX: (member_id, score_type, calculated_at DESC)

ai_branch_forecasts

Column	Type	Description
id	UUID	PK
branch_id	UUID	FK
forecast_type	VARCHAR(50)	revenue, peak_hour
forecast_date	DATE	
payload	JSONB	
created_at	TIMESTAMPTZ	

analytics_daily_snapshots

Column	Type	Description
id	UUID	PK
organization_id	UUID	FK
branch_id	UUID	FK
snapshot_date	DATE	
total_members	INT	
active_memberships	INT	
check_ins_count	INT	
revenue_amount	DECIMAL(12,2)	
new_members	INT	

expiring_soon	INT	
created_at	TIMESTAMPTZ	

UNIQUE: (branch_id, snapshot_date) or (organization_id, branch_id, snapshot_date)

SaaS billing (FitCore → gym customer)

saas_plans

Column	Type	Description
id	UUID	PK
tier	saas_plan_tier	
name	VARCHAR(100)	
max_branches	INT	
max_members	INT	
price_monthly	DECIMAL(12,2)	
features	JSONB	

saas_subscriptions

Column	Type	Description
id	UUID	PK
organization_id	UUID	FK
plan_id	UUID	FK → saas_plans
status	saas_subscription_status	
current_period_start	DATE	
current_period_end	DATE	

gateway_customer_id	VARCHAR(255)	
created_at	TIMESTAMPTZ	

Audit & reports

audit_logs

Column	Type	Description
id	UUID	PK
organization_id	UUID	FK
branch_id	UUID	FK
user_id	UUID	FK
action	VARCHAR(100)	member.update, device.enroll
entity_type	VARCHAR(50)	
entity_id	UUID	
old_values	JSONB	
new_values	JSONB	
ip_address	INET	
user_agent	TEXT	
created_at	TIMESTAMPTZ	

INDEX: (organization_id, created_at DESC)

report_jobs

Column	Type	Description
id	UUID	PK
organization_id	UUID	FK

requested_by	UUID	FK → users
report_type	VARCHAR(50)	
parameters	JSONB	date range, branch
format	VARCHAR(10)	pdf, csv, xlsx
status	VARCHAR(20)	queued, processing, done, failed
file_url	TEXT	S3
created_at	TIMESTAMPTZ	
completed_at	TIMESTAMPTZ	

Relationship summary

Parent	Child	Cardinality	FK
organizations	branches	1:N	organization_id
organizations	members	1:N	organization_id
branches	members	1:N	branch_id (primary)
users	members	1:0..1	user_id
members	memberships	1:N	member_id
membership_plans	memberships	1:N	plan_id
memberships	invoices	1:N	membership_id
invoices	payments	1:N	invoice_id
members	attendance_records	1:N	member_id
devices	attendance_records	1:N	device_id
members	device_user_mappings	1:N	member_id
trainers	member_trainers	1:N	trainer_id

Key constraints & rules

1. Every business row includes `organization_id` (except platform tables).
2. API enforces `branch_id` IN user's permitted branches.
3. `member_code` unique per organization.
4. `invoice_number` unique per organization.
5. Soft-delete members with `deleted_at`; retain financial records.
6. Dedupe device punches via (`device_id`, `external_punch_id`).
7. QR stored as `token_hash` only.

Index strategy (performance)

- `attendance_records` (`branch_id`, `check_in_at` DESC) — dashboard today
- `memberships` (`end_date`) WHERE `status` = 'active' — expiry cron
- `payments` (`payment_date`) — revenue reports
- `access_events` (`branch_id`, `occurred_at`) — security audit
- `ai_member_scores` (`member_id`, `score_type`, `calculated_at` DESC) — latest score